



Embedding Trust via Social Context in Virtual Spaces



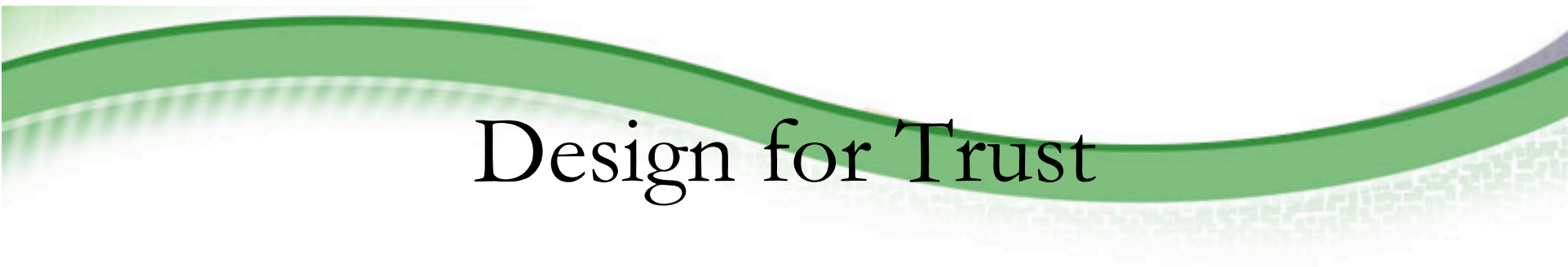
Outline

- Problem definition
- Design approach
- Proposed system
 - Interfaces
 - Simulation
 - Protocols
 - Interfaces leads protocols because design was framed by human parameters



Core Problem Statement

How to inform individual assessments of trustworthiness of a potential online transaction.



Design for Trust

- Start with human trust behaviors
- Trust
 - Used for simplification
 - Encompasses discrete technical problems
 - privacy, integrity, data security
 - Embeds discrete policy problems
 - business behavior, customer service, quality of goods, privacy



Human vs. Computer Trust

- Social science
 - Experiments to look at trust extensions
 - common assumption info sharing == trust
 - People are highly trusting
- Philosophy
 - Trust is a need
 - high default to trust
 - Trust is a tool for simplification
- Economics
 - Trust as game theoretic



Research on Humans Suggest...

- Humans may not differentiate between machines
 - We like to lump
 - Computers differentiate
- Humans become more trusting of ‘the network’
- Humans default to trust
 - Confirmed by philosophical macro observation
 - Confirmed by computer security incidents
 - Validated by fraud efficacy
 - unfortunate reams of validation



Trust & Individiation

- People interacting with a computer do not distinguish between computers as individuals but rather respond to their experience with "computers"
 - People begin too trusting
 - People learn to trust computers
 - first observed by Sproull on net in computer scientists in 1991
 - confirmed by all future privacy experiments
 - Computers are perceived as moral agents
- People will continue to extend trust - so creating another source of trust doesn't defeat trusting behaviors



Differentiation

- The tendency to differentiate between remote machines decreases with computer experience
 - More use results in more lumping
 - Explains common logon/passwords
 - along with cognitive limits
 - “My Internet is Down”
- Need explicit DO NOT TRUST signals



Observations

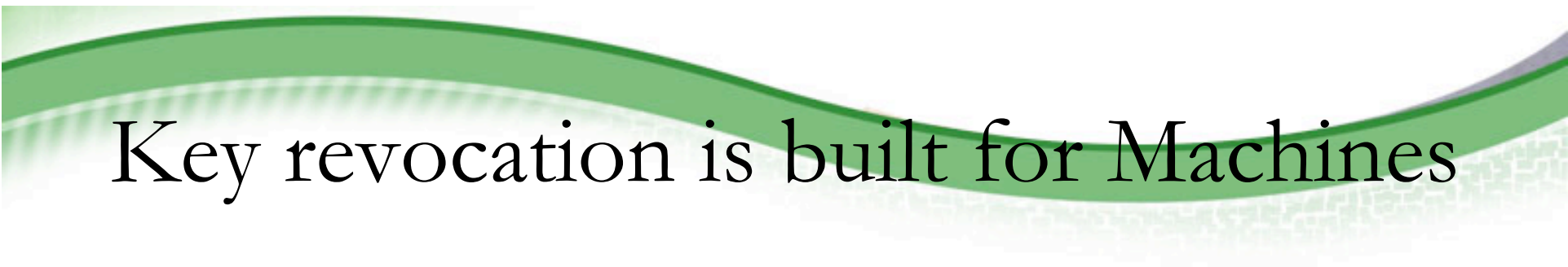
- Users are bad security managers
 - PGP, P3P, passwords,
- Security should necessarily be a default
- Surveys illustrate a continuing confusion of privacy & security
 - educate All Net Users
 - build upon this

Computer Security is Built for Machines

- Passwords
 - Humans are a bad source of entropy
- SSL
 - Two categories: secure and not secure
 - Does not encourage individiation
 - Computer security should seek to differentiate machines
 - Every site should include a unique graphic with the lock

Privacy standards are built for machines

- P3P assumes
 - All merchants trustworthy w.r.t. their own policies
 - Assumes increasingly sophisticated user
 - e.g., preference expression and negotiation
 - One standard for all transactions
- PGP
 - Monotonic increase in trust
 - No reset
 - No decrease in rate of trust extension
 - to compensate for increasing trust
 - No global or local reset
 - e.g. change in status



Key revocation is built for Machines

- CRL tend to be single level
- Different levels of revocation are needed
 - Falsified initial credential
 - all past transactions suspect
 - Change in status
 - future transactions prohibited
 - Refusal of renewal
 - current systems adequate
- CRL should reflect the entire systems in which they work, including the social system



WHAT TO DO?

- Computers
 - Process data
 - Store data
 - Transmit data
 - Distinguish
 - atomicity, privacy, availability,
- Humans
 - Understand context
 - Evaluate uncertainty
 - Make lumping decisions based on context
- Begin with the human as the basis of the design
 - Examine human interactions
 - Signal humans using pre-existing social capital



Not Even Talking to Users

- Identity theft
 - Unauthorized use of authenticating information to assert identity in the financial namespace
 - Internal process violation - Choicepoint (at least 145k records)
 - All access to the Choicepoint database was authorized
 - Subsequent misuse was authorized by the information obtained via Choicepoint
 - Security Violation - Berkeley
 - Confidentiality information - Bank of American backup data 1.2M records
- Risk profile is similar for individuals in all three cases
 - 40,000,000 credit card numbers “lost”
 - distinct risk profile

Trust and Context



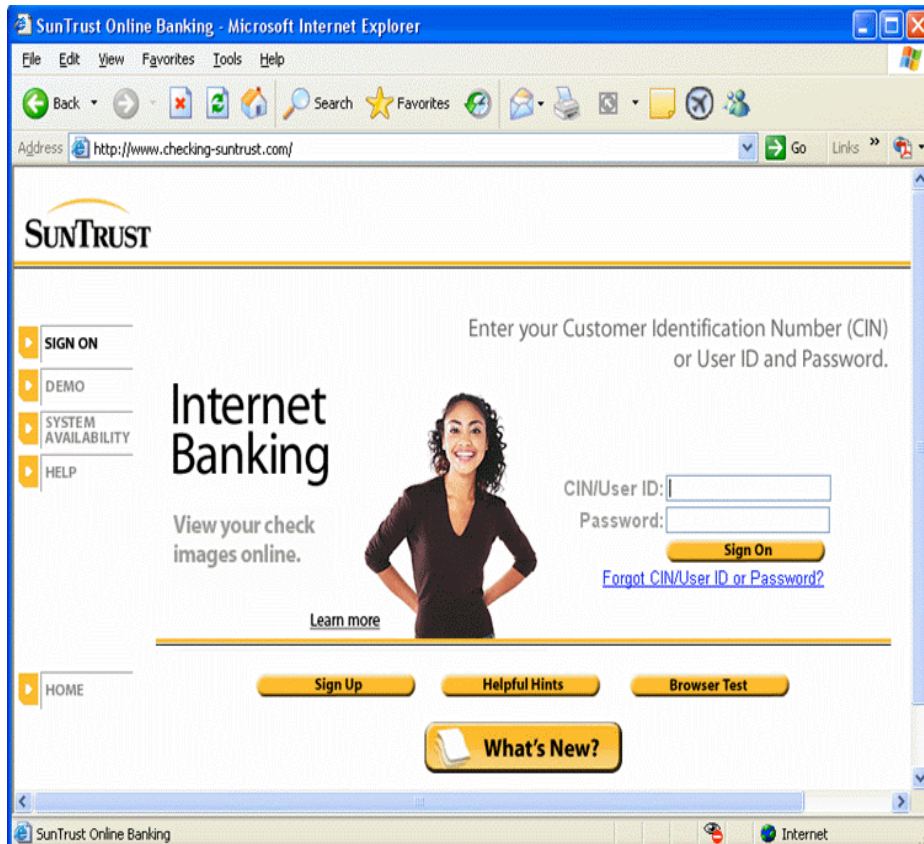
vs.



Resource Verification

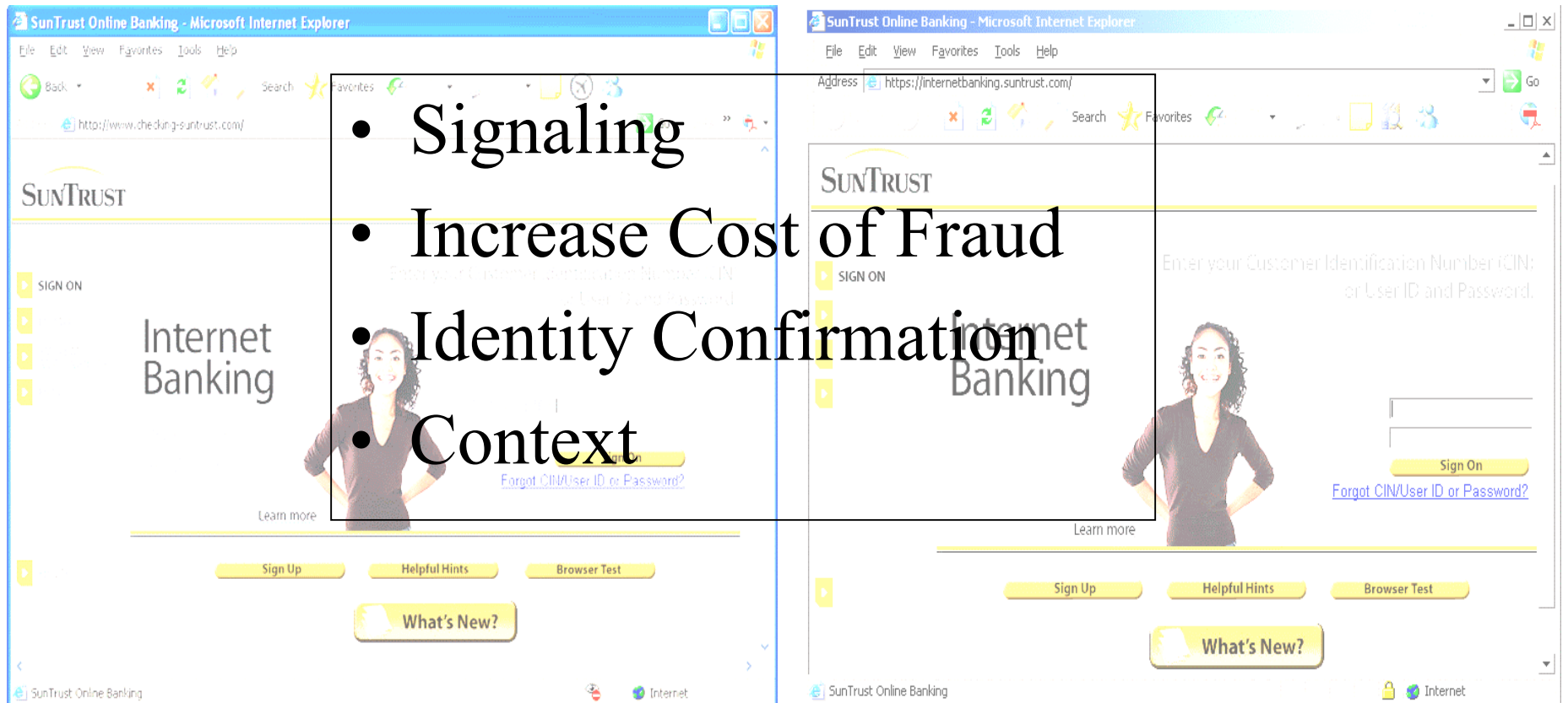
Resources are often fairly easy to identify as
“good” or “bad” in physical realms

Trust and Context



Identity Verification

Possible Solutions



Contextual information is needed for trust to reduce complexity.

Signaling

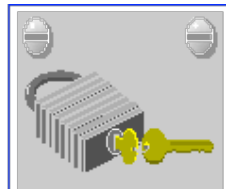


Seals



Traditional mechanisms to communicate trustworthiness.

Signaling Requires Malicious Party to Cooperate



PRIVACY
Personal Info Is Secure



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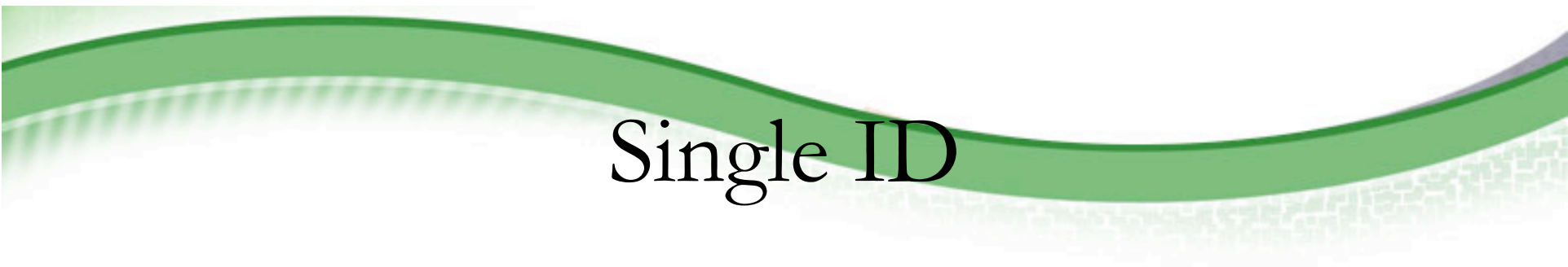
Economics Requires Payment

- Reliable payment
- New opportunities for fraud
- Perversely increase incentive?
- Setting uniform prices or negotiating prices
 - Price discrimination is the opposite of privacy
- Fraud management in a new realm



Identity Confirmation

- Trust market
 - Verisign protects you from anyone who's money they won't take
 - Matt Blaze
- Uniqueness
 - PKI
 - Joe Wilson problem



Single ID

- Impractical
- No single source of legitimate trust
- Trust behaviors vary widely within social networks and is not deterministic
 - Is Walmart a good place to shop?
 - Is Prada a good place to buy shoes?



Cradle to Grave ID.... So What

- Authentication as what? For what? By Whom?
 - Identification as having what attributes?
- Scope of namespace
 - License to drive
 - requires driving test
 - SSN
 - taxpayer ID to assert right to benefits
 - Birth certificate
 - proof of age
 - Define a credit namespace that allows for large scale illegal employment
 - Use one mechanism that applies to banking, credit, video rental, health care, ..
 - Cell phone requires that you have paid for it
 - DL requires you know how to drive



Perfect Single ID

... for every namespace

... and every context

... for all people

definitions: <http://www.ljean.com/>

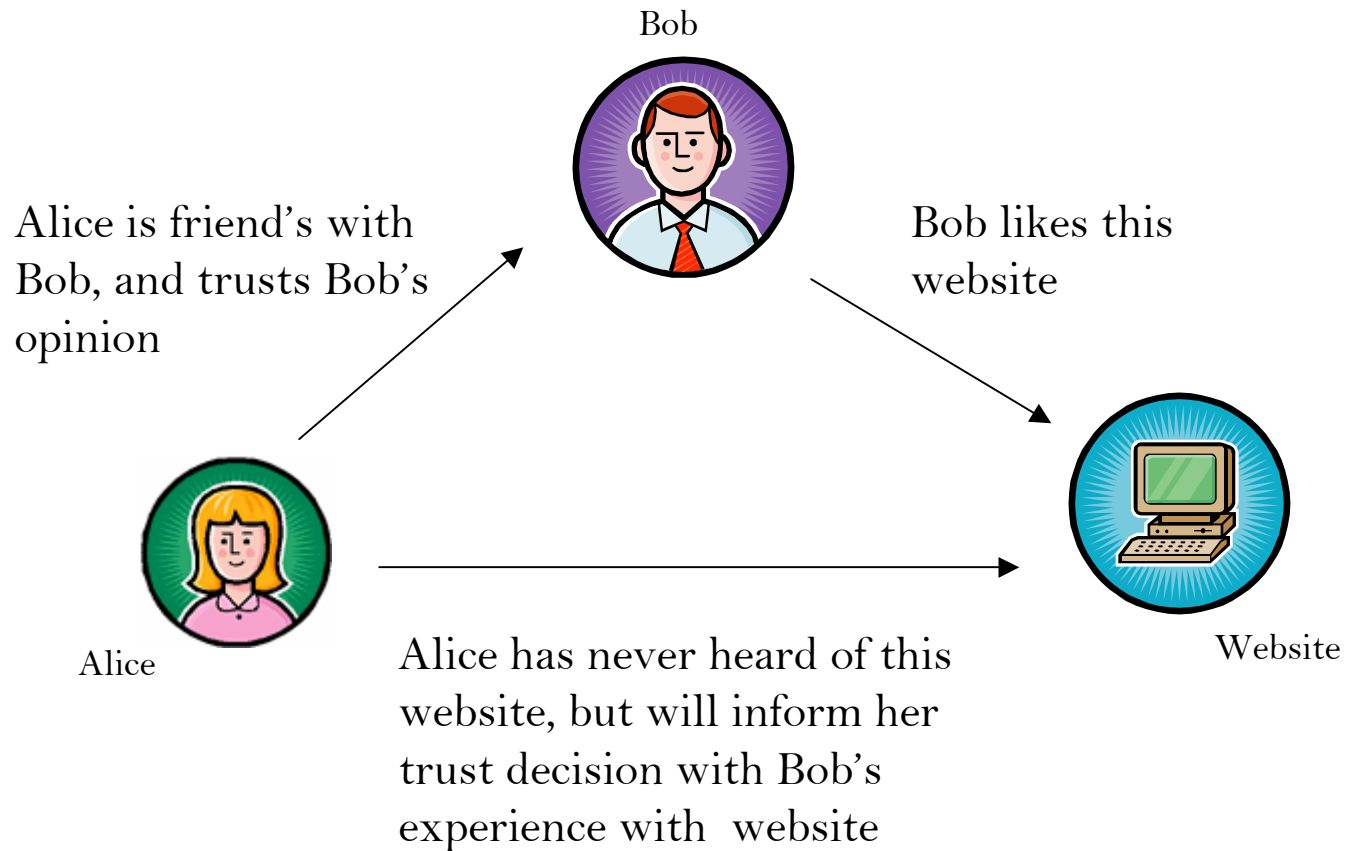


... or solve the problem at hand
by enabling contextually rational
trust behavior

Embedding Browsing in Social Context

- First trust challenge
 - Enabling trust to allow entry onto the net
 - pre {84, 87, 91} the Net was a “trust environment” because of entry barriers
 - Enabling monetary flows
- Second trust challenge
 - Providing meaning trust information
 - TrustE, BBB, Verisign
 - Namespaces for specific trust assertions
 - Christian, gay friendly, responsible merchants
 - Requires a common understanding of the limits of the namespaces
 - Transitivity
 - Automated trust decisions
 - Consistency across contexts or explicit definition of context
 - E.g., purchase a book
 - » On divorce; impotency; effective job searching; number theory

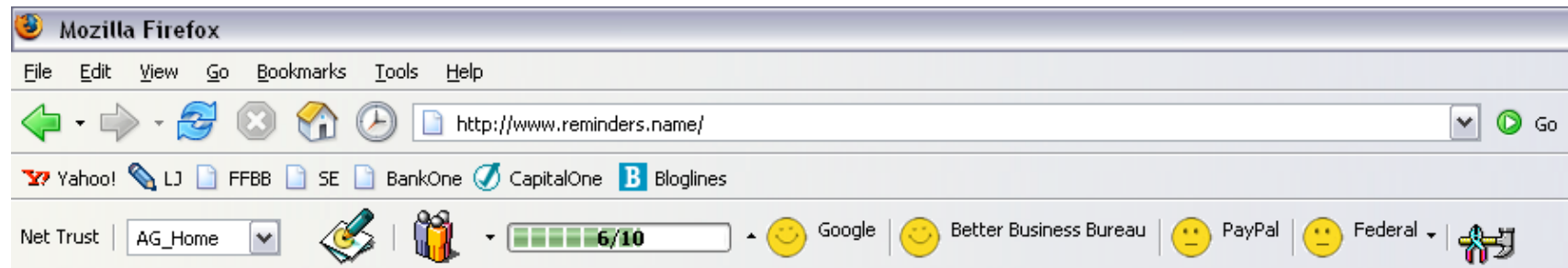
Embedding Trust via Context



Net Trust View

Using a user's **social network** (known as a buddy list) as well as user-selected **centralized authorities** (known as broadcasters) the Net Trust system displays meaningful information to the user so they can make an educated decision about the trustworthiness of a website.

The Net Trust Toolbar





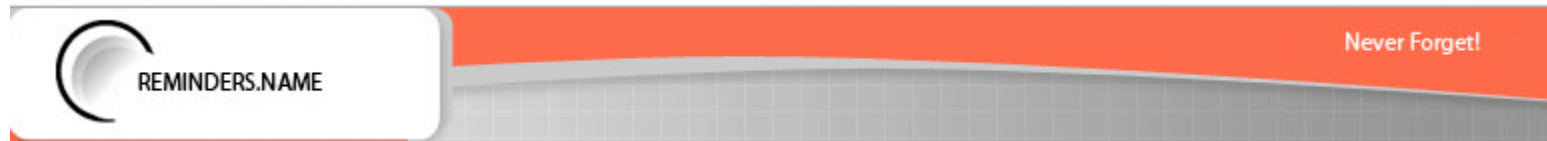
Usability

Initial Usability Testing

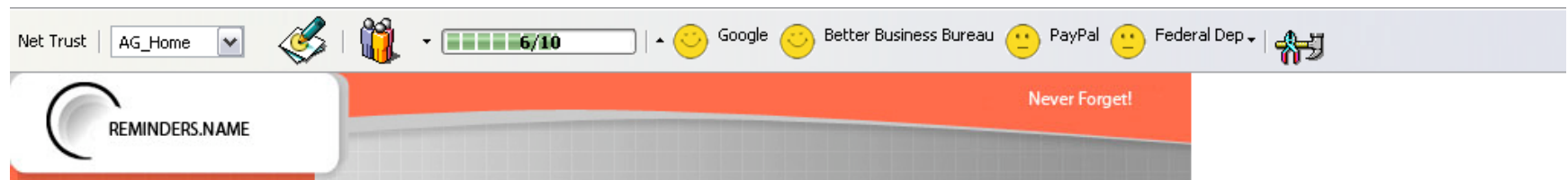
- 25 Students
- Undergraduates/Graduates
- Informatics Department, Indiana University

First Results: Reminders.name

Without Toolbar: 60% say they do not trust this site



With Toolbar: 42% say they do not trust this site

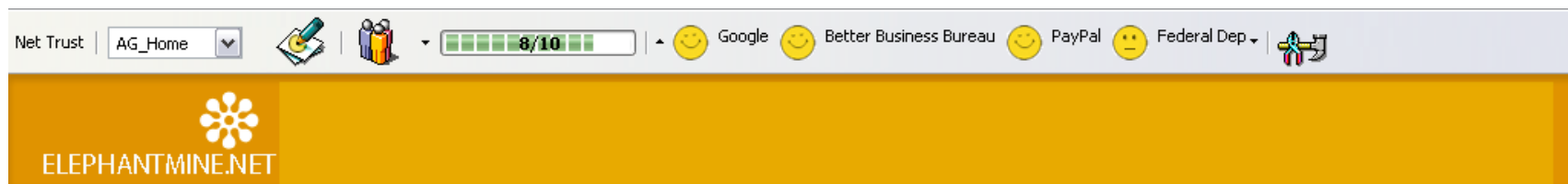


Second: Elephantmine.net

Without Toolbar: 52% say they do not trust this site



With Toolbar: 24% say they do not trust this site

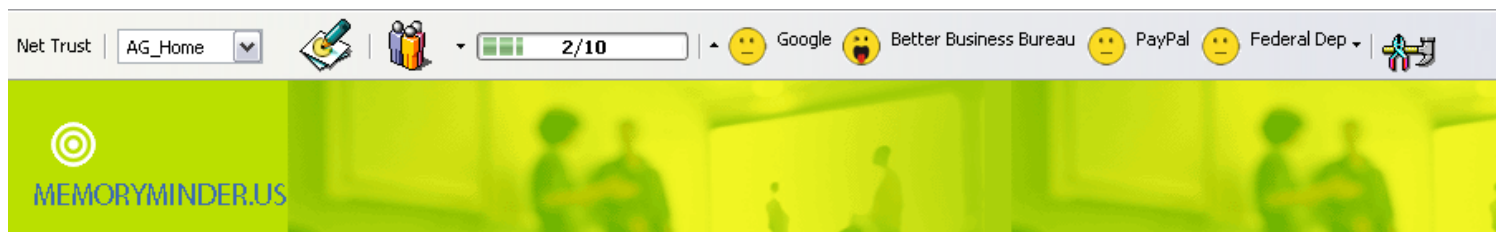


Third: MemoryMinders.us

Without Toolbar: 80% say they do not trust this site



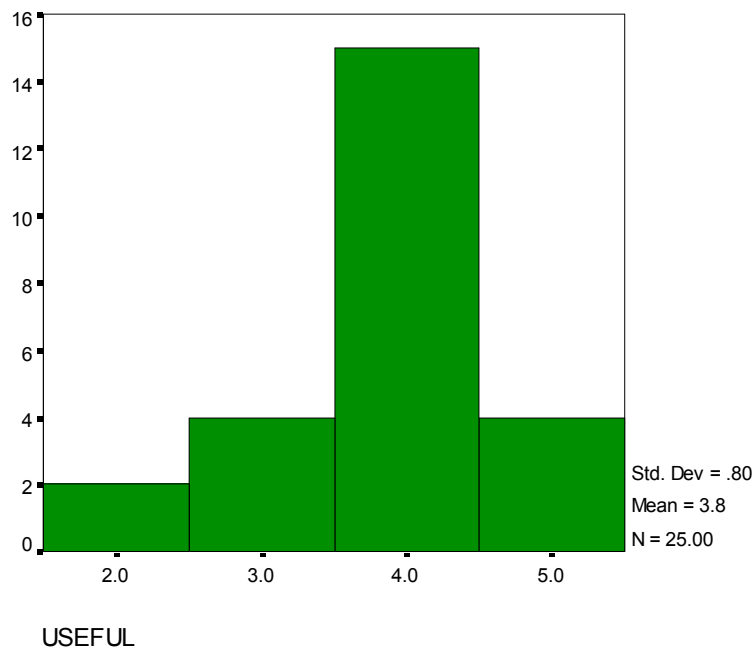
With Toolbar: 76% say they do not trust this site



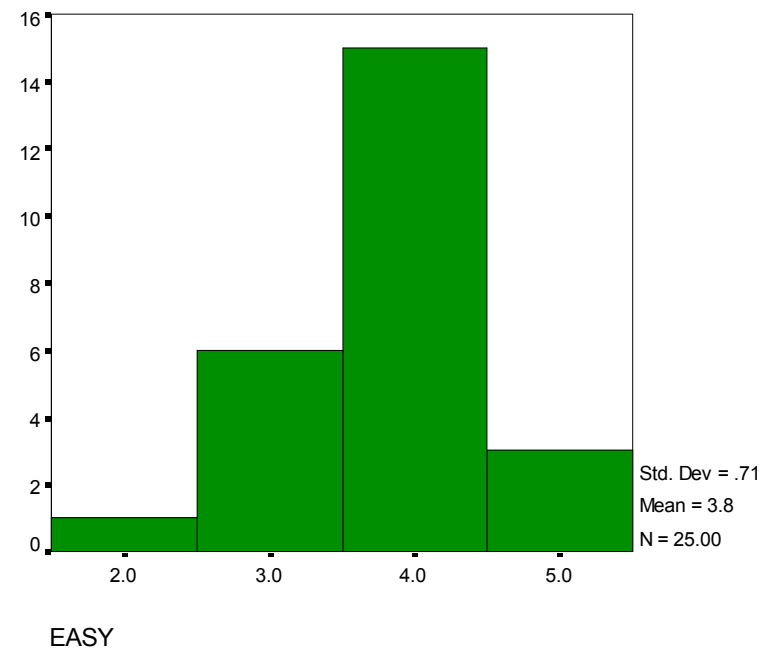
Mixed feedback does not result in statistically significant changes in user trust behavior.

Usability Results

The System is Useful:



The System is Easy to Use:





Usability Results

Finally..

80% of participants said they found the interface **MEANINGFUL**

And..

86% of participants said they would **ENJOY** using this system

Later..

Adding explicit negative peer information, communicating the null set as a negative trust signal



Abstract the Resource Problem

- Will this work in theory?
- Resources are typed as either good or bad
- Bad resources do not exhibit strategic behavior
- Good resources have some enduring identifier
- Limited ability to discern type



Fraud & Phishing Abstracted

- Resources are typed as either good or bad
- Bad resources do not exhibit strategic behavior
- Good resources have some enduring identifier
- Limited ability to discern type

Claim: when the distribution of resource availability is correlated with the distribution of users, social network ties can be leveraged to provide users with information to predict type.



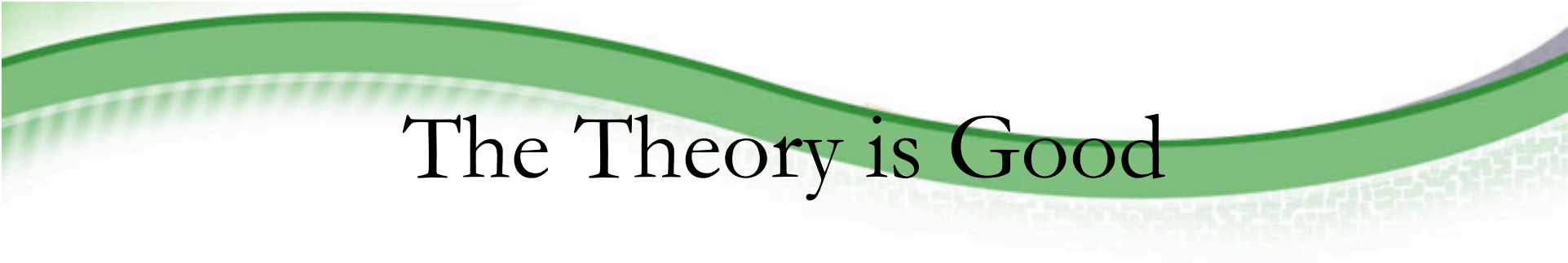
Simulation Summary

- Very simple model of networked actors deciding whether or not to visit resource
 - Network: extend Jin, Girvan & Newman (2000) to include homophily
- Decision rule: a function of number of neighbors who have also visited that resource



Simulation Results

- Under basic conditions, networked actors are very good at rejecting bad resources without avoiding good resources.
- A mechanism is needed to seed the network with good information.
- The network amplifies the power of individual detection abilities.
- Temporal signatures of bad resources (phishing) can be detected.
- BUT: non-savvy actors cannot achieve perfect (95%+) results without exogenous information sources.



The Theory is Good

- Complies and leverages human trust behaviors
- Simulations suggest potential value
- How to build it?



Problems

- Single Trusted server
 - Privacy of content can be preserved with encryption
 - Trivial traffic analysis
 - Subversion of user information
 - correlation of “identities”
 - `alice@work == alice@home == monkeygrrl@tpmcafe`
 - Single point of failure
 - obviously solvable
 - Questionable economic model



References

- “Re-Embedding Existing Social Networks into Online Experiences to Aid in Trust Assessment”, SSRN Working Paper 707139, Alla Genkina and L. Jean Camp
- “Social and Network Trust,” *DIMACS*, 14 - 15 April 2005, L Jean Camp, Alla Genkina and Allan Friedman
- L. Jean Camp, “Peer Production of Security & Privacy Information,” *Telecommunications Policy Research Conference* (Alexandria, VA).



Questions?

Thank you for your attention.